Physical Chemistry CUME, Nov. 13, 2015

Spin Matrices

An electron is a spin ½ particle. If a magnetic field is applied along the Z direction, a spin ½ particle can have two possible eigenstates with distinct energies. The two states are typically referred to as the up-spin |α> state and the down-spin |β> state. These two states are orthonormal.

With these two states, a matrix of any operator *A* can be constructed as

$$\begin{Bmatrix} <\alpha|A|\alpha> & <\alpha|A|\beta> \\ <\beta|A|\alpha> & <\beta|A|\beta> \end{Bmatrix},$$

1. When the spin along *Z* (S_Z) is measured, a value of $\frac{1}{2}\hbar$ is obtained for state |α> and a value of $-\frac{1}{2}\hbar$ is obtained for state |β> . Please write the matrix for S_Z.

2. When the total spin of an electron is measured, what do you expect to see for an electron in the |α>, |β> states, respectively.

3. Assume the operator for measuring spin along X is S_X, for spin along Y is S_Y, the raising and lower operator S_+ and S_- can be constructed as

$$\begin{aligned} S_+ &= S_X + iS_Y \\ S_- &= S_X - iS_Y \end{aligned},$$

It can be worked out that $\hbar|\alpha>=S_+|\beta>$ and $\hbar|\beta>=S_-|\alpha>$. Write the matrices for S_+ and S_-

4. Write the matrices for S_X and S_Y

5. Show that $v=\begin{pmatrix}1\\1\end{pmatrix}$ is an eigenfunction of S_X.

6. is $v=\begin{pmatrix}1\\1\end{pmatrix}$ properly normalized? If not, please normalize it.